

Theory & Practice of Concurrent Programming (COMP60007)

Coursework 2 (Theory)

The coursework should be undertaken in groups of up to three. A single submission should be made per group (you should submit a PDF document containing your solutions). Each member of a group will thus be awarded the same mark for the associated submission.

1. For each of the following justify whether it is a safety or liveness property. Additionally, for safety properties identify the “bad thing” of interest, and for liveness properties, identify the “good thing” of interest. **(20%)**
 - (a) Program P will terminate.
 - (b) Program P has no null pointer dereferences.
 - (c) Program P always returns an even number.
 - (d) Dr X will win a Turing Prize (Dr X is immortal).
 - (e) Dr Y will win a Turing Prize by the year 2030 (Dr Y is also immortal).

2. This questions is about formalising the operational semantics of a new consistency model called PSO (‘partial store ordering’). The PSO programming language is that of TSO in the lecture (including memory reads, memory writes, CAS and FAA instructions, and the memory fence `mfence`) extended with the `sfence` instruction described in Tutorial 2.

The PSO model is strictly weaker than TSO. Specifically, like TSO, PSO allows (1) write-read reordering (on different locations) and (2) `sfence`-read reordering. Additionally, unlike TSO, PSO also allows (3) *write-write reordering on different locations*. PSO admits no other instruction reordering.

- (a) Consider the P_{mp} program below:

$$\begin{array}{l} x := 1; \\ y := 1; \end{array} \parallel \begin{array}{l} a := y; \\ b := x; \end{array}$$

Is there a PSO-outcome $(S, -)$ of P_{mp} with $S(\tau_2)(a) = 1$ and $S(\tau_2)(b) = 0$? Justify this informally in terms of possible instruction reorderings and interleavings. **(10%)**

- (b) Consider the P'_{mp} program below:

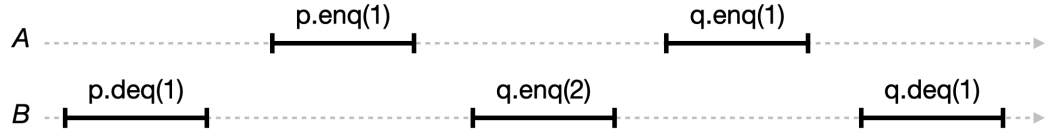
$$\begin{array}{l} x := 1; \\ \text{sfence}; \\ y := 1; \end{array} \parallel \begin{array}{l} a := y; \\ b := x; \end{array}$$

Is there a PSO-outcome $(S, -)$ of P'_{mp} with $S(\tau_2)(a) = 1$ and $S(\tau_2)(b) = 0$? Justify this informally in terms of possible instruction reorderings and interleavings. **(10%)**

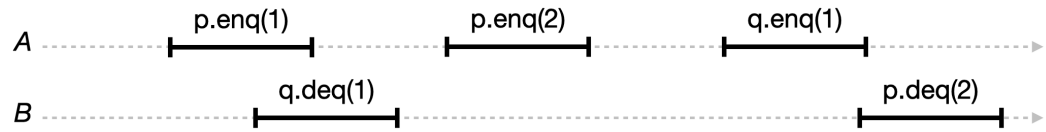
- (c) The sequential and program transitions of PSO are identical to those of TSO. The *memory transitions* of PSO are somewhat different but can be described analogously to TSO with transitions of the form $M, B \xrightarrow{\tau:l}_m M', B'$. Write down the memory transitions of PSO. That is, write the PSO memory transitions for executing a memory read, a memory write, a successful RMW, a failed RMW, a memory fence, a store fence, and for removing the entries from the thread buffers. **(40%)**

3. In all the histories below p and q are queue objects where ‘enq’ denotes the enqueue method, ‘deq’ denotes the dequeue method and ‘size’ denotes the size method (returning the number of entries in the queue).

(a) Is the history H shown below i) linearizable? ii) sequentially consistent? (10%)



(b) Is the history H' shown below sequentially consistent? (5%)



(c) Is the history H'' shown below linearizable? (5%)

